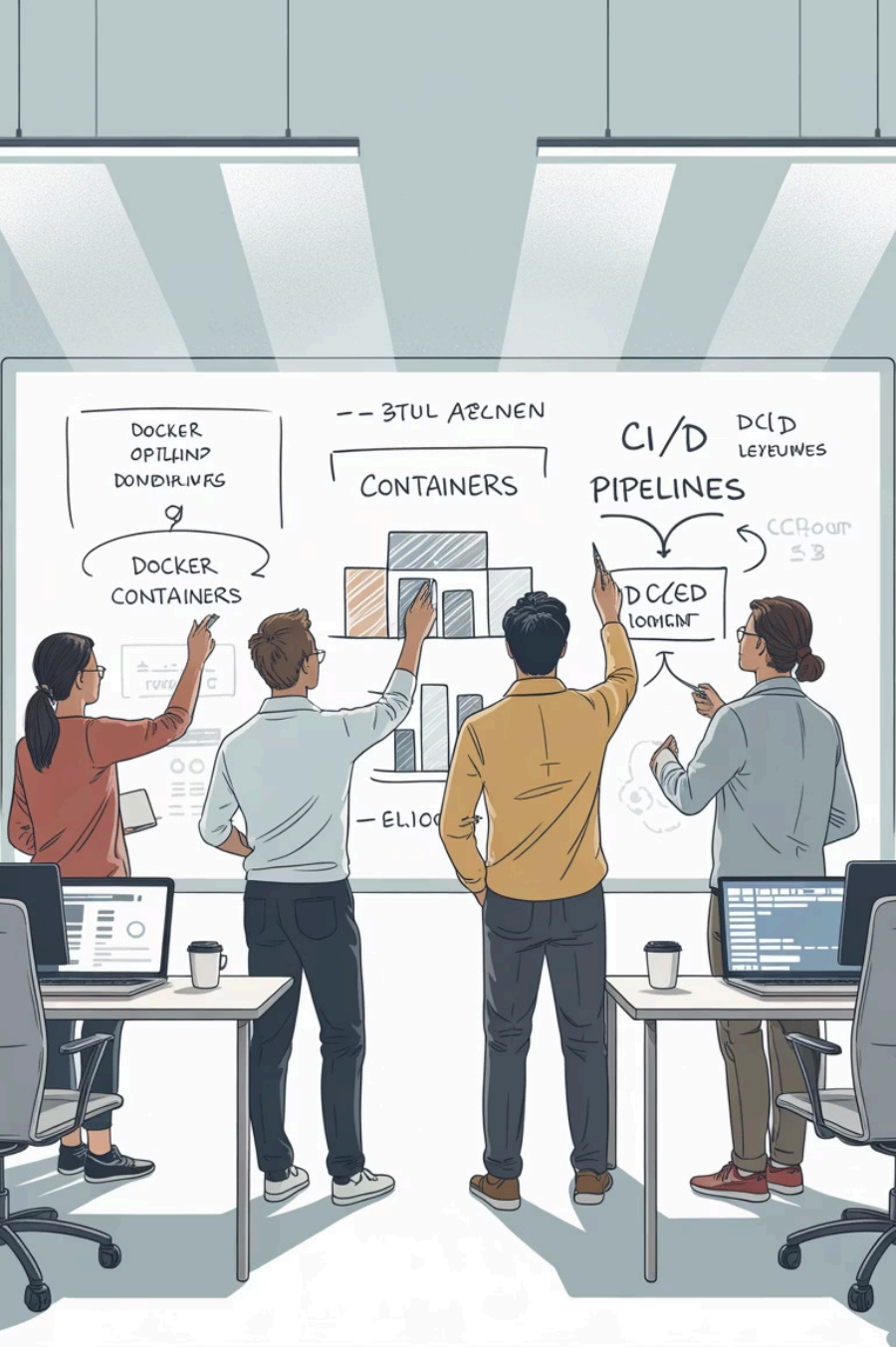




Introduction to DevOps & Docker



Title Slide

DevOps & Docker: Modern Software Development Made Simple



Presented by: Naresh Chaurasia

? Why DevOps?

Problems in traditional development:



Slow releases



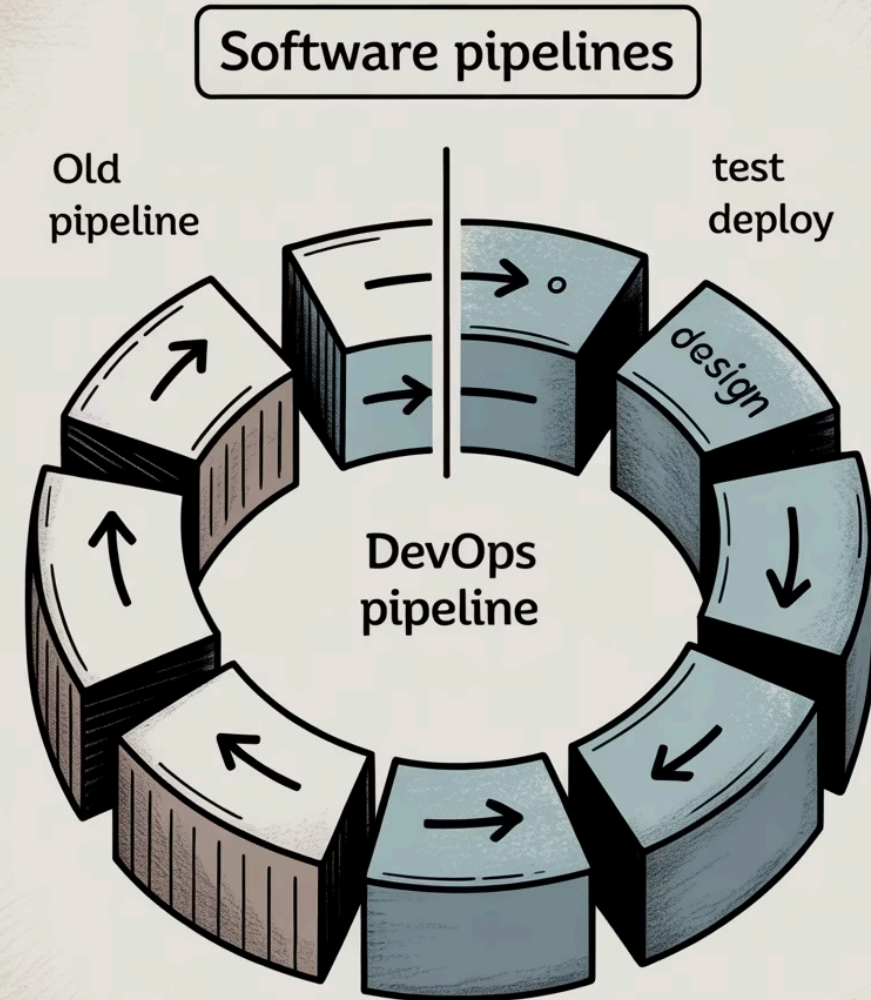
Miscommunication between
Dev & Ops



Manual deployments = more errors

Need → Faster, reliable, collaborative approach

Visual: Old vs. New software pipeline diagram





What is DevOps?

 Combination of *Development + Operations*

 Not just tools → *Culture + Practices + Tools*

 **Goal** → Faster delivery, better quality, continuous improvement

Visual: DevOps infinity loop

DevOps Lifecycle

 Continuous Integration (CI)

 Continuous Monitoring



 Continuous Delivery (CD)

 Continuous Testing

Visual: Infinity loop labelled with these stages

Key Benefits of DevOps



Faster releases



Higher quality software



Reduced downtime



Better collaboration between teams

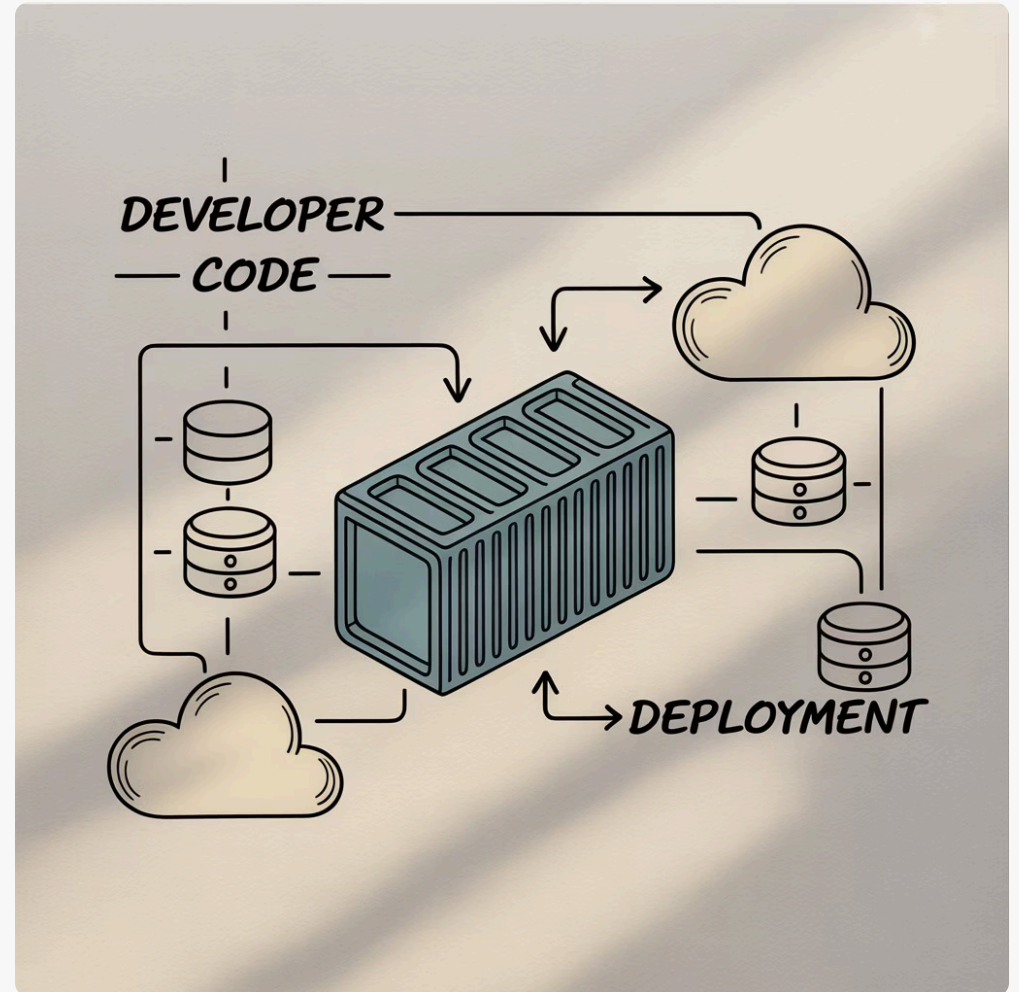
Where Docker Fits In?

Problem: "It works on my machine!" 🤔

Solution: *Containers ensure same environment everywhere*

Docker = most popular containerisation platform

Visual: Developer → Docker container → Cloud





What is Docker?



Lightweight *container platform*



Packages app + dependencies into one container



Runs anywhere ( laptop,  server,  cloud)

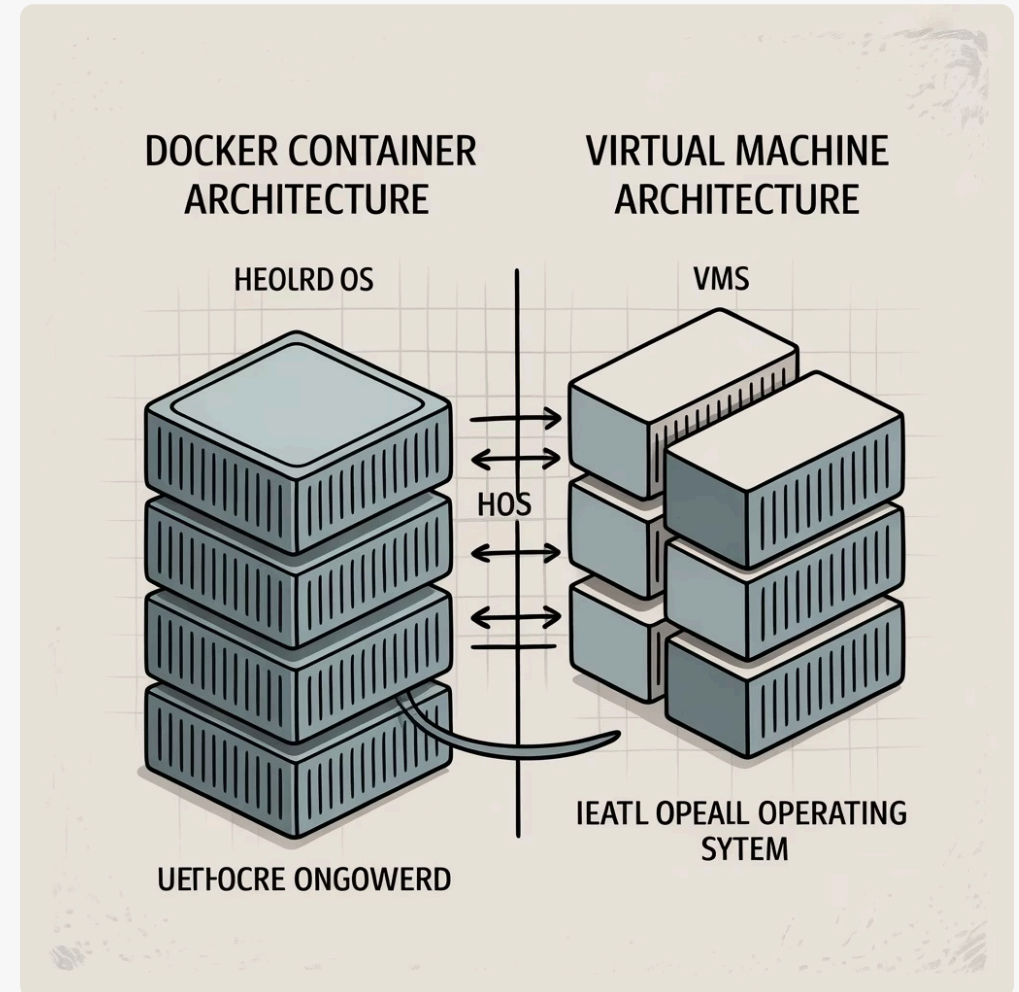
Docker vs Virtual Machine (VM)

VM: Heavy, needs full OS 🐌

Docker: Lightweight, shares host OS ⚡

Result: Faster startup, less resource usage

Visual: Side-by-side Docker vs VM architecture diagram



Docker Workflow in DevOps



 Developer writes code



 Build Docker image



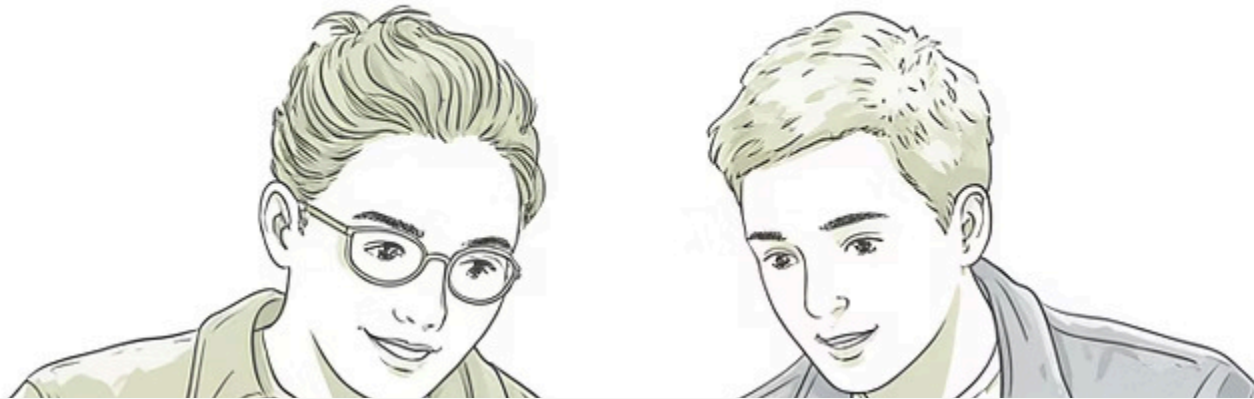
 Push to Docker Hub / registry



 Deploy container to servers/cloud



 Scale easily with Kubernetes



Real-World Example

College project: web app with multiple team members

✗ Without Docker → "Works on my laptop only"

✓ With Docker → Runs the same everywhere

Career Relevance



DevOps + Docker = 🔥 high demand skills



Used in startups, enterprises, cloud computing



 Boosts employability

Recap & Key Takeaways

DevOps = *culture + tools* for faster delivery

Docker = *key DevOps tool* for containers

Together → Modern, scalable, reliable software



Q&A / Thank You

Encourage questions from students

Resources:

-  docker.com
-  YouTube tutorials on DevOps basics
-  Free courses on CI/CD pipelines